

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA5101

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 5

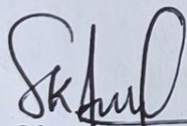
Total Marks:50

Annexure A

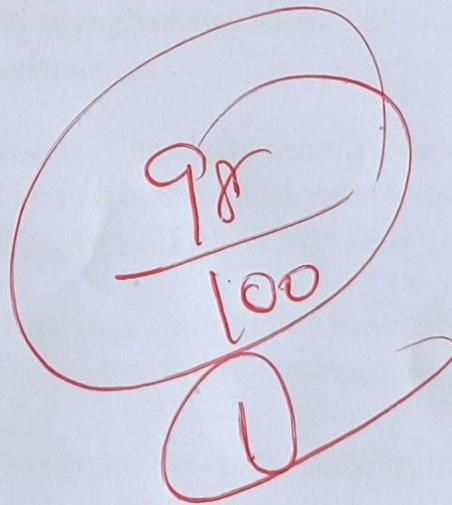
APPLICATION- BASED QUESTIONS

Code of Conduct:

I Shaik Asha (192473025) Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student:



98
100

APPLICATION- BASED QUESTIONS

1. Scenario: Event Communication Security: A student organization is preparing to share confidential event details with its core members using a simple classical encryption method. During testing, a message "MEET AT NOON" is transformed into an encoded version using a fixed shift pattern.

- a. Reconstruct how the encrypted message might appear if a shift of 4 positions is applied.
- b. Explain how an authorized member, aware of the encoding rule, would retrieve the original message.

2. Scenario: Intercepted Suspicious Message: A cybersecurity intern intercepts the coded message: "GSRH RH Z HVXIVG" during network monitoring. The structure of the text suggests a simple substitution mechanism rather than complex encryption.

- a. Deduce the most likely original message.
- b. Justify your answer by analysing repeating patterns, letter structures, or known cipher characteristics.

3. Scenario: Secure Messaging System: A company deploys a keyword-based encryption technique for internal communication. An employee sends the word "HELLO" using the keyword "KEY" to ensure confidentiality.

- a. Demonstrate how the encryption process transforms the message step-by-step.
- b. Explain how the repeating keyword influences each character of the ciphertext.

4. Scenario: Suspicious Digital Asset Investigation: During a digital forensic audit, investigators suspect that a company logo file may contain hidden confidential information embedded within it. The file appears normal upon casual inspection.

- a. Analyze how investigators could detect the presence of hidden data within the image.
- b. Recommend two techniques that could strengthen the security of such hidden communications against detection or extraction.

5. Scenario: Legacy Encryption in Corporate Communication: A legacy system in a company still uses a transposition-based encryption method with three levels of arrangement. A received message appears as: "WECRLTEERDSOEEFEAOCAIVDEN".

- a. Determine the most probable original message by reversing the encryption process.
- b. Critically evaluate why such a method is vulnerable in modern communication systems.
- c. Propose one practical enhancement to improve the confidentiality of messages in this scenario.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25 23	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30 30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20 20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15 15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10 10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

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Scenario: INTERCEPTED SUSPICIOUS MESSAGE

Given Ciphertext,

"GSRH RH Z HVXINGI"

The structure of the ciphertext suggests a simple substitution cipher. The most likely cipher is the Atbash cipher.

a. Deduce the most likely Original Message

Answer:

The original message is: "THIS IS A SECRET"

The cipher is Atbash, a type of monoalphabetic substitution cipher.

In Atbash, each alphabet character is replaced by its corresponding character from the reverse alphabet.

Atbash Substitution Table:

Plaintext	A	B	C	D	E	F	G	H	I
Ciphertext	Z	Y	X	W	V	U	T	S	R
Plaintext	J	K	L	M	N	O	P	Q	R
Ciphertext	Q	P	O	N	M	L	K	J	I
Plaintext	S	T	U	V	W	X	Y	Z	
Ciphertext	H	G	F	E	D	C	B	A	

Character-by-character decoding:

G	S	R	H		R	H		Z		H	V	X	I	V	G
↓	↓	↓	↓		↓	↓		↓		↓	↓	↓	↓	↓	↓
T	H	I	S		I	S		A		S	E	C	R	E	T

Hence, GSRH RH Z HVX I V G → THIS IS A SECRET

b. Justification

The answer can be justified using word patterns, letter substitution, and known cipher characteristics.

1. Word-length analysis

The cipher text has 4 words

GSRH	RH	Z	HVX I V G
4	2	1	6

The proposed plain text has exactly the same structure

THIS	IS	A	SECRET
4	2	1	6

Substitution ciphers preserve the no. of characters and word boundaries.

2. One-letter word

The ciphertext contains 'Z'.

Common one-letter English words are A and I.

Under Atbash: Z → A

This provides strong evidence for the plaintext word A.

3. Consistent Substitution

In a substitution cipher, one ciphertext character consistently represents one plaintext character.

Ex: $G \rightarrow T$

$S \rightarrow H$

$R \rightarrow I$

$H \rightarrow S$

$GSRH \rightarrow THIS$

The mapping remains consistent throughout the message.

4. Reverse alphabet characteristic

Atbash specifically uses the reverse alphabet.

All the mappings in the ciphertext follow the reverse-alphabet relationship.

5. Meaningful plaintext

After substitution:

$GSRH RH Z HVXIVG \rightarrow THIS IS A SECRET$

The resulting sentence is grammatically correct & meaningful.

Cipher Characteristics:

Feature	Atbash Cipher
Type	Substitution cipher
Category	Monoalphabetic
Key mechanism	Reverse alphabet
Word Spaces	Preserved
Letter Positions	Preserved

Security Observation: Atbash is not suitable for protecting

sensitive modern communications because it uses a fixed substitution pattern. An attacker who identifies the cipher can easily decrypt the message.

Conclusion:

The intercepted ciphertext: GSRH RH Z HVXVGI is an Atbash cipher.

The decoded message is "THIS IS A SECRET"

The deduction is supported by word lengths, one-letter word analysis, consistent substitution, reverse-alphabet characteristics, meaningful plaintext, and successful encryption verification.

2. Scenario: SUSPICIOUS DIGIT ASSET INVESTIGATION

A company logo appears normal, but investigators suspect that confidential information has been embedded inside it. This is a possible case of image steganography.

a. How can investigation detect hidden data?

1. Understand the suspected technique

Steganography is the process of hiding secret information inside an apparently normal file.

For an image:

Cover Image + Secret Data



Steganographic Embedding



Stego Image

The stego image should appear normal to an ordinary viewer.
For example, information can be hidden by modifying the Least Significant Bits (LSBs) of image pixels.

2. Preserve the evidence

Before investigation, investigators should create a forensic copy of the image.

Suspicious image



Preserve Original



Create Forensic Copy



Perform Analysis

The original file should not be unnecessarily modified.

A cryptographic hash such as SHA-256 can be calculated.

Image



SHA-256



Unique Hash Value

This helps verify the integrity of the evidence.

3. File & Metadata Analysis

Examine: → File format & size

Tool: ExifTool

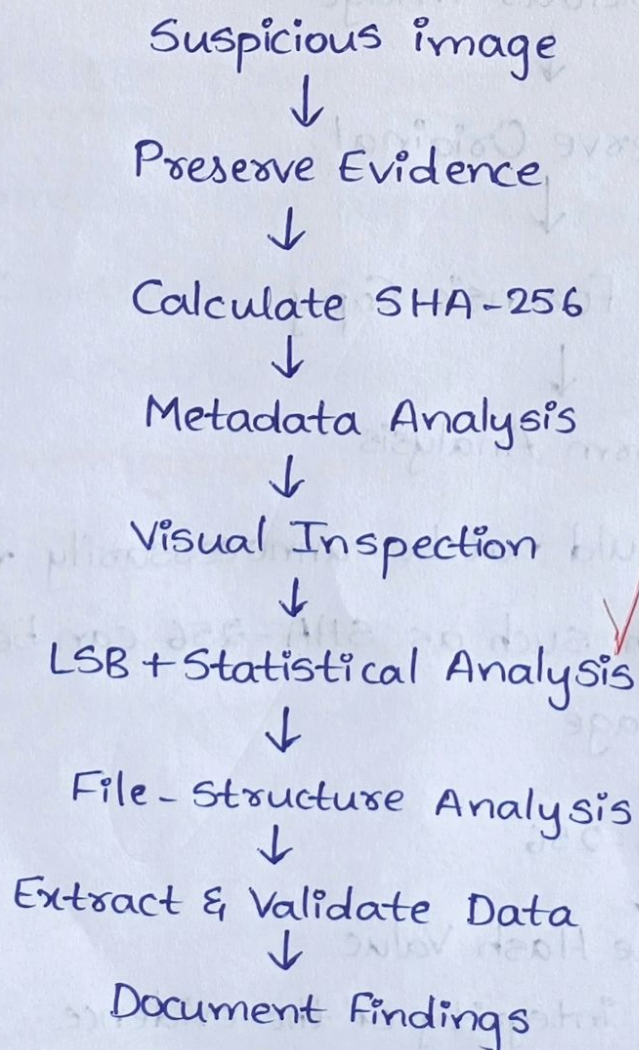
- Image Dimensions & color depth
- Compression Details
- Creation/modification dates
- Metadata & embedded comments
- Software information

4. Visual Inspection

Compare the suspicious image with a known original & look for

- unusual colors/noise
- pixel distortions
- strange borders/regions
- unexpected visual differences

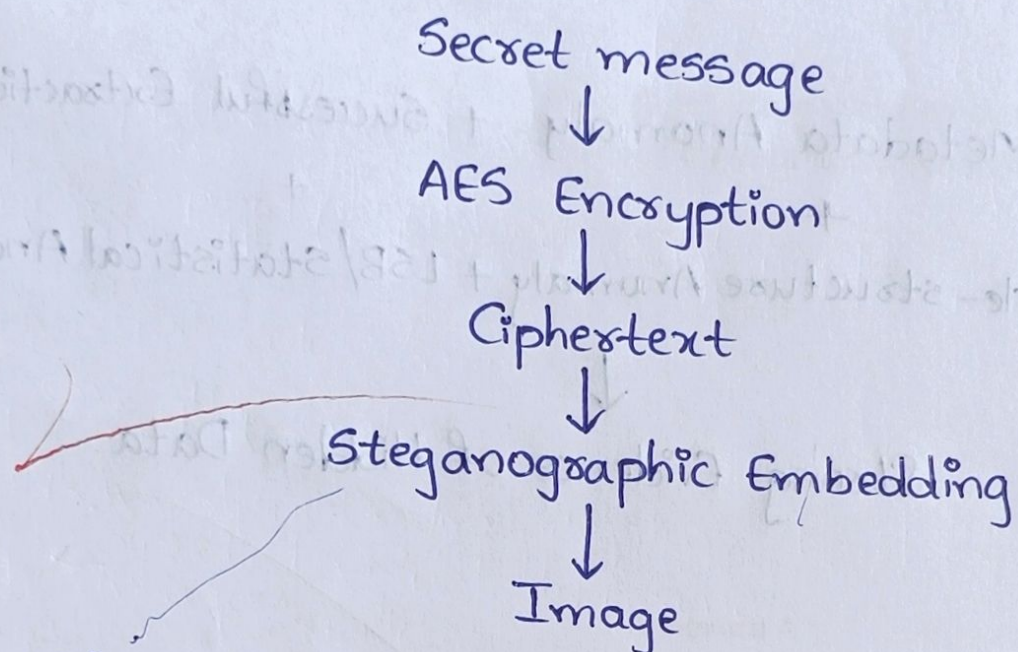
Investigation Workflow:



b. Two techniques to strengthen Hidden Communication

1. Encrypt Before Embedding

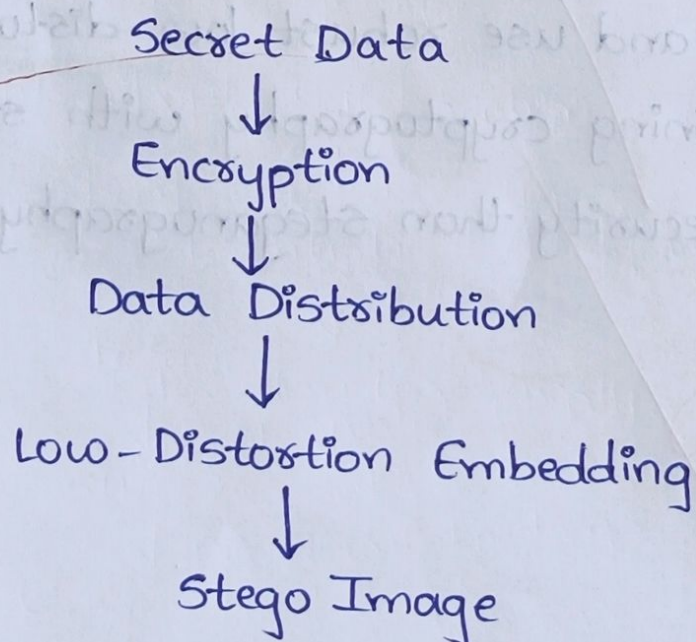
The secret message should be first encrypted using a strong algorithm such as AES.



Even if an attacker extracts the hidden data, it will remain unreadable without the encryption key.

2. Use Robust, Low-Distortion Steganography

Use a steganographic strategy that distributes hidden data carefully & minimizes changes to the image.



Advantages: 1. Reduces visible distortion

2. Makes statistical detection harder

3. Makes direct extraction more difficult

Analysis: A single indicator does not prove the presence of hidden data.

Metadata Anomaly + Successful Extraction
+
File-structure Anomaly + LSB/Statistical Anomaly



Strong Evidence of Hidden Data.

Conclusion:

Hidden data in a company logo can be detected using metadata analysis, visual inspection, LSB analysis, statistical steganalysis, and file-structure analysis. Tools such as ExifTool, Binwalk, zsteg, and StegExpose can assist the investigation.

For stronger protection, encrypt the secret before embedding it and use robust, low-distortion steganography. Combining cryptography with steganography provides stronger security than steganography alone.

